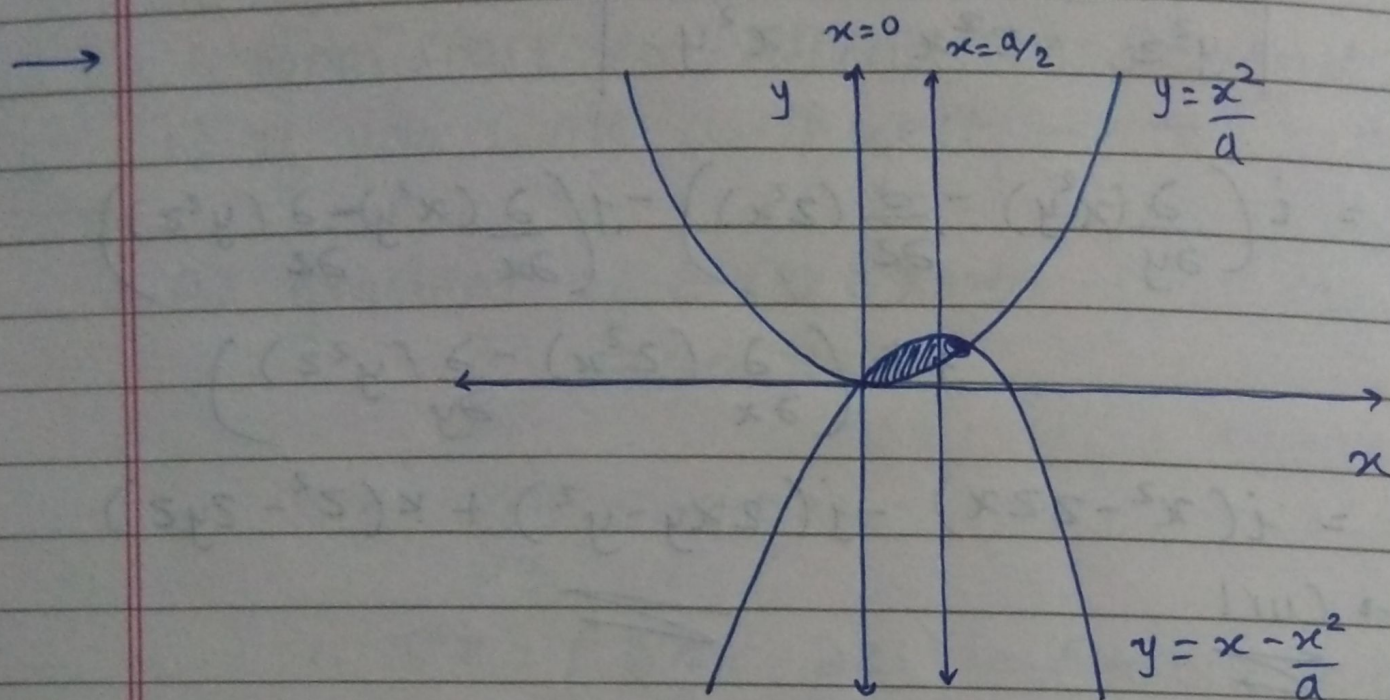


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A] Sketch following regions.

13] Region bounded by $x=0$, $x=\frac{a}{2}$, $y=\frac{x^2}{a}$ & $y=x-\frac{x^2}{a}$



B] Evaluate the following.

4] $\iint_R e^{y^2} dx dy$ where R = triangular region with vertices $(0,0)$, $(2,1)$ & $(1,1)$.

→ $A(0,0)$, $B(2,1)$ & $C(1,1)$

From $A(0,0)$ to $B(2,1)$,

$$\text{Slope} = \frac{1-0}{2-0} = \frac{1}{2} //$$

& From $A(0,0)$ to $C(1,1)$,

$$\text{Slope} = \frac{1-0}{1-0} = 1 //$$

& From $B(2,1)$ to $C(1,1)$,

$$\text{Slope} = \frac{2-1}{1-1}$$

From $B(2,1)$ to $C(1,1)$, this is horizontal line at $y=1$ from $x=1$ to $x=2$.

for y ranging from 0 to 1

left boundary $\Rightarrow x = 2y$ (from line AB)

right boundary $\Rightarrow x = y$ (from line AC).

$$\iint_R e^{y^2} dx dy = \int_0^1 \int_{2y}^y e^{y^2} dx dy$$

$$\therefore \int_{2y}^y e^{y^2} dx = e^{y^2} (y - 2y) \\ = -ye^{y^2}$$

$$\therefore \int_0^1 -ye^{y^2} dy$$

let $u = y^2$

$$\therefore du = 2y dy \text{ or } dy = \frac{du}{2y}, \text{ when } y=0, u=0 \text{ \&}$$

when $y=1, u=1$.

$$\therefore -y dy = -\frac{1}{2} du$$

$$\therefore -\frac{1}{2} \int_0^1 e^u du$$

$$\therefore -\frac{1}{2} [e^u]_0^1 = -\frac{1}{2} (e^1 - e^0) = -\frac{1}{2} (e - 1)$$

c) Check order of following & hence evaluate

$$3) \int_0^1 \int_x^{1/x} \frac{y}{(1+x)^2 (1+y^2)} dx dy$$

y ranging from 0 to 1.

& x ranging from y to $\frac{1}{y}$

$\therefore$ line $x=y$ is diagonal line in I Quadrant.

The curve $x = \frac{1}{y}$ is hyperbola that approaches axes as y inc.

intersectⁿ pts,

$$y = \frac{1}{y}$$

$$\therefore y^2 = 1 \quad \therefore y = 1 \quad (\text{since } y \geq 0)$$

$$\therefore \text{intersect}^n \text{ pt} \Rightarrow (1, 1).$$

For x , it ranges from 0 to 1

For each fixed x , y ranges from x to $\frac{1}{x}$

$$\therefore \int_0^1 \int_x^{1/x} \frac{y}{(1+x)^2(1+y^2)} dy dx$$

$$\therefore \int_x^{1/x} \frac{y}{1+y^2} dy$$

Substitute, $u = 1+y^2 \quad \therefore du = 2y dy \quad \therefore dy = \frac{du}{2y}$
when $y = x, u = 1+x^2$

$$y = \frac{1}{x}, u = 1 + \frac{1}{x^2}$$

$$\therefore \frac{1}{2} \int_{1+x^2}^{1+\frac{1}{x^2}} \frac{1}{u} du = \frac{1}{2} [\ln(u)]_{1+x^2}^{1+\frac{1}{x^2}} = \frac{1}{2} \left(\ln\left(1 + \frac{1}{x^2}\right) - \ln(1+x^2) \right)$$

$$\therefore \frac{1}{2} \ln\left(\frac{1+\frac{1}{x^2}}{1+x^2}\right) = \frac{1}{2} \ln\left(\frac{1+x^2}{x^2(1+x^2)}\right)$$

$$= \frac{1}{2} \ln\left(\frac{1}{x^2}\right) = -\ln(x)$$

$$\therefore \int_0^1 \frac{-\ln(x)}{(1+x)^2} dx$$

$$\therefore \int_0^1 -\ln(x) dx = 1.$$

$$= 1$$

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Q7] Solve following.

10] Find area region betⁿ ellipses $\frac{x^2}{4} + \frac{y^2}{9} = 1$ & $\frac{x^2}{1} + \frac{y^2}{4} = 1$

→ Ellipse 1 :-

semi-major axis, $a = 3$ (y-axis)

semi-minor axis, $b = 2$ (x-axis)

Ellipse 2 :-

semi-major axis, $a = 2$ (y-axis)

semi-minor axis, $b = 1$ (x-axis)

$$\therefore A = \int_{-1}^1 (y_1(x) - y_2(x)) dx$$

$y_1(x)$ = upper boundary. - Ellipse 1.

$y_2(x)$ = lower boundary - Ellipse 2.

$$\therefore y_1 = 3 \sqrt{1 - \frac{x^2}{4}} \quad (\text{upper half}).$$

$$\therefore y_2 = 2 \sqrt{1 - x^2} \quad (\text{upper half}).$$

$$\therefore A = 2 \int_0^1 \left(3 \sqrt{1 - \frac{x^2}{4}} - 2 \sqrt{1 - x^2} \right) dx$$

$$\therefore \int_0^1 3 \sqrt{1 - \frac{x^2}{4}} dx$$

$$\text{let } u = \frac{x}{2} \therefore dx = 2du.$$

$$\therefore \Rightarrow 6 \int_0^{1/2} \sqrt{1 - u^2} du = 6 \cdot \frac{\pi}{8} = \frac{3\pi}{4} //$$

$$\therefore \int_0^1 2 \sqrt{1 - x^2} dx = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2} //$$

$$\therefore A = 2 \left(\frac{3\pi}{4} - \frac{\pi}{2} \right) = 2 \left(\frac{3\pi - 2\pi}{4} \right) = \frac{\pi}{2} //$$